FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS – GRADE 10 | SUB-STRAND 1.1: REAL NUMBERS	
Student Assessment Document	
Student Name	______________________________________
Class	______________________________________
Date	______________________________________

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation document. Your goal is to write a complete, well-organised answer to the driving question using everything you have learned across all 8 lessons. Read the driving question carefully, then use the sections below as a guide to build your answer. Each section has a prompt telling you what to write about and an exemplar showing you the standard expected. Write in your own words — do not copy the exemplar. Use mathematical vocabulary, real-life Kenyan examples, and refer back to the Ugali Pot Problem whenever it helps you explain your thinking. Your final answer should feel like one connected piece of writing, not a list of facts.

Section 1 – The Problem with the Sufuria: Setting Up the Driving Question

Begin your explanation by describing the Ugali Pot Problem in your own words. Why is the circumference of the sufuria puzzling, even though the diameter is a perfectly whole number (28 cm)? What mathematical question does this situation force us to ask about numbers? Make sure you introduce the idea that not all numbers behave the same way.	When a mama mboga in Gikomba Market multiplies the diameter of her aluminium sufuria — exactly 28 cm — by π to find the circumference, her calculator displays 87.96459430… and the digits continue without end and without any repeating pattern. This is puzzling because 28 is a clean whole number, yet the circumference it produces cannot be written as an exact decimal or a simple fraction. When her daughter cuts the string to 88 cm, there is always a tiny gap or overlap — proof that 88 is only an approximation. This everyday situation forces a deeper mathematical question: why can some numbers be written down completely and exactly, while others seem to go on forever? To answer this, we need to understand the full system of real numbers and the fundamental difference between rational and irrational numbers.

Section 2 – Mapping the Real Number System: From Counting Numbers to Irrationals

Explain the structure of the real number system. Describe each subset — natural numbers, whole numbers, integers,	The real number system is a family of numbers that are nested inside one another, each set growing larger as we include more types of numbers. Natural numbers (1, 2, 3, …) are the counting numbers a trader uses to count bags of unga at Gikomba Market. Whole numbers add zero to this set — useful when a maize field in the Rift Valley records zero rainfall on a dry day. Integers extend the set further to include negative numbers, such as –3°C, which could represent a temperature reading at the top of Mount

rational numbers, and irrational numbers — and explain how they are related (use the idea of a nested set or number line if it helps). Give at least one Kenyan real-life example for each subset. Then explain where π fits in this system and why.	Kenya. Rational numbers include all integers but also every fraction and decimal that either terminates or repeats — for example, the price of sukuma wiki at Ksh 12.50 (which equals 25/2) or the repeating decimal 0.333… (which equals 1/3). Irrational numbers, however, cannot be written as any fraction p/q where p and q are integers and q ≠ 0. Their decimal expansions go on forever without repeating. The number π is irrational — no fraction can capture it exactly. √2 is another example. Together, rational and irrational numbers form the complete set of real numbers, which can all be placed on the number line.

Section 3 – Why π Cannot Be Written Exactly: The Nature of Irrational Numbers	
Explain what makes a number irrational. Why is it impossible to write π as a fraction or a terminating/repeating decimal? Use the sufuria example to anchor your explanation. Also give at least one other example of an irrational number (such as √2 or √3) and connect it to a real-life Kenyan context — for example, a farmer measuring a square plot of land or a builder in Kisumu calculating a diagonal length.	A rational number can always be expressed in the form p/q, where p and q are integers and q is not zero. When you divide p by q, the decimal either stops (terminates) or falls into a repeating cycle of digits. For example, 3/4 = 0.75 (terminates) and 2/3 = 0.666… (repeats). Irrational numbers obey no such pattern — their decimal digits continue forever in a sequence that never repeats and never ends. π is irrational because mathematicians have proven that no fraction of two integers can equal it exactly. The calculator value 87.96459430… for the sufuria's circumference reflects this: every extra decimal place gets us closer to the truth but never reaches it. Another example is √2 ≈ 1.41421356…, which arises when a farmer in Nakuru digs a square plot with sides of exactly 1 metre and wants to know the length of the diagonal — the answer is √2 metres, a number that cannot be written as a fraction. This is not a fault of our calculators or our measuring tools; it is a fundamental property of these numbers themselves.

Section 4 – Working Accurately with Numbers That Go On Forever: Rounding, Significant Figures, and Surds	
Explain the strategies that mathematicians and Kenyan professionals use to work accurately with irrational numbers in real life. Discuss: (a) rounding and significant figures — when is this good enough and what error does it introduce? (b) leaving answers in surd form or in terms of π — why is this sometimes more exact and more useful? Give specific examples from contexts such as construction, cooking, healthcare, athletics,	Because irrational numbers cannot be written completely, we use two main strategies depending on how much precision we need. The first strategy is rounding. When the mama mboga's daughter cuts a string to 88 cm (rounding 87.96459430… to the nearest centimetre), she introduces an error of about 0.04 cm — small enough for a piece of string but potentially significant in engineering. A construction engineer in Nairobi building a circular water tank would round to more decimal places, perhaps using 87.9646 cm, to reduce the error. Using π ≈ 3.14 or π ≈ 22/7 are common approximations — note that 22/7 is rational and is only an approximation of π, not equal to it. The second strategy is to work in exact form: leaving answers as surds (like √3 or 2√5) or in terms of π (like 28π cm). A civil engineer designing a roundabout in Kisumu would write the exact circumference as 28π metres and only convert to a decimal at the final step of their calculation. This avoids rounding errors building up across many steps. Kenyan athletes, too, benefit from precision — a 400-metre track is designed using exact π calculations so that each lane has a fair running distance. Both strategies — rounding appropriately and working in exact form — are essential tools for anyone who works with real numbers professionally.

or farming in Kenya.	

Section 5 – Answering the Driving Question: Why Some Numbers Go On Forever and How We Handle It	
Now bring everything together to answer the driving question directly and completely. Explain: (1) why some everyday numbers can be written exactly (they are rational) while others cannot (they are irrational); (2) how the real number system organises all these numbers; and (3) how mathematicians and Kenyan professionals work accurately with numbers that never end. Finish with a reflection — why does it matter that we understand this distinction? What would go wrong if we treated every approximation as exact?	We can never write some everyday numbers down completely because they are irrational — their decimal expansions go on forever without forming a repeating pattern, and no fraction of two integers can represent them exactly. The number π is the most famous example: the diameter of a sufuria can be a perfect whole number, 28 cm, yet the circumference it produces is 28π — a value that exists on the number line but cannot be captured in full by any finite sequence of digits. Other measurements — the price of unga at Ksh 60.50, the distance from Nairobi to Nakuru at 160 km — are rational numbers: they can be expressed as fractions (121/2 and 160/1 respectively) whose decimals terminate or repeat. The real number system organises all these numbers into a hierarchy: natural numbers sit inside whole numbers, which sit inside integers, which sit inside rational numbers; alongside rational numbers — but not overlapping — lie the irrational numbers; and together they fill every point on the number line to form the complete set of real numbers. Mathematicians and Kenyan professionals handle irrational numbers in two powerful ways: they round to an appropriate number of decimal places when a practical measurement is needed, accepting a small, known error; or they work in exact surd or π form throughout a calculation, converting to a decimal only at the final step to avoid accumulating rounding errors. Understanding this distinction matters enormously. If the mama mboga's daughter believed 88 cm was the exact circumference rather than an approximation, she might not question the gap in her string. If an engineer in Mombasa treated π as exactly 3.14 when designing a bridge arch, the accumulated error across thousands of calculations could affect safety. Knowing which numbers are rational, which are irrational, and how to work with both accurately is not just a mathematical exercise — it is the foundation of precise, honest, and safe professional practice.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of the Real Number System	Accurately and completely describes all subsets of the real number system (natural, whole, integer, rational, irrational) with correct relationships between them. Consistently uses precise mathematical vocabulary. Provides multiple relevant Kenyan examples for different subsets and correctly places π and other irrationals within the system.	Describes most subsets of the real number system with correct relationships. Uses mathematical vocabulary mostly correctly. Provides at least one Kenyan example per subset and correctly identifies π as irrational.	Describes some subsets but with gaps or errors in how they relate (e.g., confuses integers and whole numbers, or incorrectly classifies rational numbers). Limited or no Kenyan examples. May incorrectly classify π or other numbers.
Explanation of Rational vs. Irrational Numbers	Clearly and accurately defines rational numbers (p/q form, terminating or repeating decimals) and irrational numbers (non-terminating, non-repeating). Provides a convincing explanation of why π cannot be written as a fraction. Distinguishes between approximations like 22/7 and the true	Correctly defines both rational and irrational numbers and explains the key difference. Explains why π is irrational in general terms. References the sufuria context. May not distinguish clearly between 22/7 and π.	Shows partial understanding — may define one type correctly but not the other, or confuse 'long' decimals with irrational numbers. Explanation of why π is irrational is missing or inaccurate. Limited connection to the sufuria context.

	value of π. Uses the sufuria context effectively throughout.		
Strategies for Working with Irrational Numbers	Clearly explains both strategies — rounding (with awareness of the error it introduces) and exact surd/π form — and justifies when each is appropriate. Provides specific, realistic Kenyan professional contexts (e.g., construction, engineering, athletics, farming) showing deep understanding of why precision matters.	Explains rounding and exact form as strategies and gives at least one Kenyan professional context for each. Acknowledges that rounding introduces error. Explanation is mostly clear but may lack detail on when each strategy is most appropriate.	Mentions rounding but does not discuss exact surd or π form, or vice versa. May not acknowledge that rounding introduces error. Kenyan professional contexts are absent or vague. Limited understanding of why precision matters.
Coherent Answer to the Driving Question	The final section directly and completely answers all three parts of the driving question (why some numbers can't be written exactly; how the real number system is organised; how professionals work accurately with infinite decimals). Ties all lesson phases together into a unified, logical argument. Reflection demonstrates genuine insight into the real-world consequences of ignoring the rational/irrational distinction.	Answers at least two of the three parts of the driving question clearly. Makes reasonable connections across lessons. Includes a reflection on why the distinction matters, though it may be brief or rely on a single example.	Attempts to answer the driving question but misses one or more key parts. The response reads as a list of facts rather than a connected argument. Reflection is absent or superficial (e.g., 'it is important to know maths').
Use of Mathematical Language and Kenyan Context	Consistently uses correct mathematical terminology (irrational, surd, rational, real number, approximation, terminating, non-repeating, etc.). Seamlessly integrates specific Kenyan contexts (Gikomba Market, Nairobi–Nakuru distance, Rift Valley farming, Kisumu construction, Kenyan athletics) to ground abstract concepts in familiar, relevant situations throughout the entire response.	Uses most mathematical terms correctly with only minor errors or omissions. References Kenya-specific contexts in most sections, making the explanation mostly grounded and relatable.	Uses some mathematical vocabulary but with frequent errors or over-reliance on everyday language in place of correct terms. Kenyan contexts are largely absent or replaced with generic examples, making the explanation feel disconnected from the anchoring phenomenon.